

Plasma 2025 (Schedule)							
Day-1 (27-12-2025, Saturday) at LHC, IIT Tirupati							
Time	Registration					Room	
8:00 AM						LHC Ground Floor	
8:45 AM to 10:00 AM	Inauguration					LHC-CR-211	
10:00 AM to 11:00 AM (15+15+10+10+10 (buffer) (60 Minutes)	Session-1 Sponsor presentation session (Session Chair: Prof. A K Das) Ozone Crop Innovate Private Limited (OC IPL) (15 Mins) ATOS Instruments Marketing Services (15 Mins)					LHC-CR-211	
11:00 AM to 11:20 PM	Group Photo followed by High-Tea Break					LHC Ground Floor	
Time	Session - 2A Next-Generation Tokamak Systems Session Chair: Prof. S K Dutta, Outstanding Scientist, Director, MTRDC, DRDO	Lecture Recording	Room	Session-2	Session - 2B Plasma for Sustainable Technology Session Chair: Prof. Ram Prakash, IIT Jodhpur	Lecture Recording	Room
11:20 AM to 1:05 PM (30+30+15+15+10+05 (Buffer)) (100 Minutes)	Talk 1: Prof. Avinash Khare, Vice Chancellor, Sikkim University (30 Mins) Title: A new type of divertor for tokamak	YES	LHC-CR-101		Talk 1: Prof. Peter Bruggeman, University of Minnesota, USA (30 Mins) Title: Nonthermal Plasma Applications	YES	LHC-CR-211
	Talk 2: Prof. Joydeep Ghosh, IPR Gandhinagar (30 Mins) Title: Nuclear Fusion: From dream to reality!	YES			Talk 2: Prof. Shihabudheen M. Maliyekkal, IIT Tirupati (30 Mins) Title: Hybrid Plasma Technologies for Advanced Wastewater Treatment	YES	
	Oral 1: FST 1, Kishore Mishra, First Plasma Commissioning and Operational Highlights from India's First Spherical Tokamak at IPR (12+3 Mins)				Oral 3: PTA 1, P S Naga Sai Srikar, Characterization and application of atmospheric pressure plasmas by integrating the optical spectroscopy measurements with generalized population-kinetic models and machine learning methods (12 Mins + 3 Min)		
	Oral 2: FST 2, Maya Padivattath, Comparison of System Codes for Fusion Power Plant Design: SARAS and PROCESS (12 Mins + 3 Min)				Oral 4: PTA 2, Tejashwi Rana, Optimization of CO2 Thermal Arc Plasmas Using Graphite Electrodes: Pathways Toward Sustainable Plasma Gasification (12 Mins + 3 Min)		
	Sponsor Slot: Eltech Engineers Pvt. Ltd. (10 Mins)				Sponsor Slot: SIMCO Global Technology & System Ltd. (10 Mins)		
1:05 PM to 2:00 PM	Lunch Break					LHC Ground Floor	
Time	Session - 3A Lasr & Fuson Plasma Session Chair: Prof. Punit Kumar, Lucknow University	Lecture Recording	Room	Session-3	Session - 3B Fundamentals of Plasmas Session Chair: Prof. Arun Tangirala	Lecture Recording	Room
2:00 PM to 3:50 PM (30+30+15+15+15+05 (Buffer)) (110 Minutes)	Talk 1: Prof. Amita Das, Laser Fusion Plasma (30 Mins) Title: Novel regime of Laser Plasma Interaction (Laser Fusion Plasma)	YES	LHC-CR-101		Talk 1: Prof. M. V. Kartikeyan, Director, IIITDM Kanchipuram, India (30 Mins) Title: Gyrotrons for Plasma Heating and Other Applications	YES	LHC-CR-211
	Talk 2: Prof. Mainak Bandyopadhyay, IPR Gandhinagar (30 Mins) Title: Negative Ion Source Plasma and Its Extracted Beam: Research Activities and Opportunities at IPR	YES			Talk 2: Prof. Sudeep Bhattacharjee, IIT Kanpur (30 Mins) Title: Thermodynamics of plasmas confined by a dipole magnet	YES	
	Oral 5: FST 3, Prachi jain, Development of the TRADE System Code for Tokamak Reactor Design (12 Mins + 3 Min)				Oral 8: FPD 1, Anuravi Sharma, Magnetic field control of dc self bias and plasma parameters in capacitively coupled rf discharges (12 Mins + 3 Min)		
	Oral 6: FST 4, Geetha D, Investigation of low energy electron collision with BH3 (12 Mins + 3 Min Discussion)				Oral 9: FPD 2, Prakash Sarika Shrivastav, Stimulated Raman Side Scattering in Expanding Plasma with Thermal Effect: A Fluid Study (12 Mins + 3 Min)		
	Oral 7: TMS 1, Vishwa Bandhu Pathak, Laser wakefield acceleration studies for high-energy density light-particle collisions to observe SF-QED (12 Mins + 3 Min)				Oral 10: FPD 3, Deepak Rajputohit, Study of spatial structure and conversion efficiency of higher harmonics in whistler mode excitation in magnetized plasma of lypd (12 Mins + 3 Min)		
3:50 PM to 5:15 PM	Poster Session-I ASP_P_01 to ASP_P_36 + FPD_P_01 to FPD_P_75 + TMS_P_01 to TMS_P_10 (Total 121) (With a Tea Break in Parallel)					LHC Ground Floor	
Time	Session - 4A AMP data, Modelling and Simulation Session Chair: Prof. P C Deshmukh, IIT Tirupati	Lecture Recording	Room	Session-4	Session - 4B: Plasma Dynamics and Applications Session Chair: Prof. Rajesh Srivastava, IIT Roorkee	Lecture Recording	Room
5:15 PM to 7:35 PM (30+30+30+15+15+15+05 (Buffer)) (140 Minutes)	Talk 1: Prof. B.K. Sahoo, PRL Ahmedabad (30 Mins) Title: What is in the choice of a many-body method	YES	LHC-CR-101		Talk 1: Dr. Sushil Kumar Singh, Czech Tec. University, Prague (30 Mins) Title: Experimental Study of Laser-Plasma Interaction Using Sub-nanosecond Kilo-Joule Class Laser	YES	LHC-CR-211
	Talk 2: Prof. Animesh Kuley, IISc Bengaluru (30 Mins) Title: Global Gyrokinetic simulation of fusion Plasmas	YES			Talk 2: Dr. Apeksha Madhukar, IIT Goa (30 Mins) Title: Flexible surface Dielectric Barrier Discharge for Next-gen plasma application	YES	
	Talk 3: Dr. Santanu Karmakar, MTRDC Bengaluru (30 Mins) Title: Gyrotron and its applications for plasma heating	YES			Oral 16: FPD 5 Suryasunil Rath, Machine Learning with Interpretability for Predicting Plasma Parameters in Microwave Atmospheric Plasmas (12 Mins + 3 Min)		
	Oral 11: TMS 2 Shanila M, Machine Learning Enhanced Optical Diagnostics of Atmospheric Pressure Non-Thermal Air Plasmas (12 Mins + 3 Min Discussion)				Oral 14: PTA 3, Renu Bahl, Role of Convergent–Divergent Magnetic Nozzle Profiles in Enhancing Plasma Flow Acceleration in Helicon Plasma Thruster (12 Mins + 3 Min)		
	Oral 12: TMS 3: Sripan Mondal, Numerical Simulation of Oscillatory Reconnection Controlled by Alternating Current-driven Waves in Laboratory Plasma (12 Mins + 3 Min Discussion)				Oral 15: FPD 4 Satish Babu Thota, Diagnostics of Ar Atmospheric Pressure Plasma Jet using Time-resolved Optical Emission Spectroscopy and Imaging Measurements (12 Mins + 3 Min)		
	Oral 13: TMS 4, Mohammad Yasir, Discussions on Electron Heating and Energy Absorption in Plasmas via Vortex Beams (12 Mins + 3 Min Discussion)				Talk 3: Prof. Ahmad Hamdan, Université de Montréal, Canada (30 Mins) Title: Dynamics of nanosecond pulsed discharges close to a liquid surface (Online Talk)	YES	
7:35 PM to 9:00 PM	Dinner					LHC Ground Floor	
Day-2 (28-12-2025, Sunday) at LHC, IIT Tirupati							
Time	Session - 5A High-Temperature and Fuson-Relevant Plasmas Session Chair: Dr. Raj Singh, IPR, Gandhinagar	Lecture Recording	Room	Session-5	Session - 5B Plasma for Space Science & Technology Session Chair: Prof. D N Gupta, University of Delhi	Lecture Recording	Room
9:00 AM to 11: 20 AM (30+30+30+15+15+15+05 (Buffer)) (140Minutes)	Talk 1: Dr. Srikumar Ghorui, BARC, Mumbai (30 Mins) Title: A Unique Non-Divergent Plasma Beam for Plasma Material Interaction Under Fusion Relevant Conditions: A BARC Development	YES	LHC-CR-101		Talk 1: Mr. K. M. Shanhbogue, Dy. Director, LPSC and Program Director, EPS, ISRO (30 Mins) Title: Plasma Thrusters for Space Science and Technology Applications	YES	LHC-CR-211
	Talk 2: Dr. J. Padma Nilaya, BARC, Mumbai (30 Mins) Title: Plasma Research at BARC: Present and Future	YES			Talk 2: Prof. Umesh Kadhane, IIST Thiruvananthapuram (30 Mins) Title: Plasma-induced processes in the evolution of organics in space	YES	
	Talk 3: Prof. K Ramchandran Bhartiyar University (30 Mins) Title: Thermal Plasma Simulations	YES			Oral 20: CDT 1, Vajitha G, Unveiling Plasma-Assisted Activation of Biomass-Derived Carbons for Superior Electrochemical Applications (12 Mins + 3 Min Discussion)		
	Oral 17: ASP 1, Manthan Verma, Scaling and Intermittency In Magnetohydrodynamics Turbulence (12 Mins + 3 Min)				Oral 21: CDT 2 Nani Medicherla, Design of Ka band Metamaterial Abasorber for Defence Applications (12 Mins + 3 Min Discussion)		
	Oral 18: ASP 2 Ravinder Bhambhu, Radiation Pressure driven Rayleigh- Taylor Instability in Compressible Ultra-relativistic Degenerate Magnetized Dense Plasma (12 Mins + 3 Min)				Oral 22: CDT 3 Sanjay Chandra Andola, Characterization of a Carbon Fiber Hybrid X-Pinch in the Microsecond Current Regime (12 Mins + 3 Min)		
	Oral 19: ASP 3 Pratik Patel, Ion-Cyclotron and Whistler Waves in Three-dimensional Hall Magnetohydrodynamic Turbulence (12 Mins + 3 Min Discussion)				Oral 23: CDT 4 Ramesh Patel, A Miniaturized End-Fire Circularly Polarized Antenna Based on Spoof Surface Plasmon Polaritons (12 Mins + 3 Min)		
11:20 AM to 1:00 PM	Poster Session-II: FST_P_01 to FST_P_82 + CDT_P_01 to CDT_P_28 + TMS_P_11 to TMS_P_20 (Total 120) (With a Tea Break in Parallel)					LHC Ground Floor	
1:00 PM to 2:00 PM	Lunch Break						
Time	Session - 6A Exotic & Dusty Plasma Session Chair: Prof. Swatantra Pratap Singh, IIT Bombay	Lecture Recording	Room	Session-6	Session - 6B Quantum computing & AI/ML in Plasma Session Chair: Prof. S Aravinda, IIT Tirupati	Lecture Recording	Room
2:00 PM to 4:05 PM (30+30+30+15+15+05 (Buffer)) (125 Minutes)	Talk 1: Dr. Prashant K Singh, TIFR Hyderabad (30 Mins) Title: Compact, femtosecond laser driven plasma accelerator	YES	LHC-CR-101		Talk 1: Dr. Srinivasa Prasanna V, TCG CREST-CQuERE, Kolkata (30 Mins) Title: Quantum algorithms and computation	YES	LHC-CR-211
	Talk 2: Dr. Surabhi Jaiswal, IISER Pune (30 Mins) Title: Revisiting plasma crystal and structural transition phenomena	YES			Talk 2: Prof. P. K. Karmakar, Tezpur University (30 Mins) Title: Normal acoustic modes in degenerate quantum plasmas	YES	
	Talk 3: Prof. Sanat Tiwari, IIT Jammu (30 Mins) Title: Dusty plasma turbulence in DC discharge plasma	YES			Talk 3: Prof. Jan Trieschmann, Kiel University, Germany (30 Mins) Title: Data-Integrated Physical Simulation and Machine Learning Analysis: The Case of Plasma Processing of SiO x /Cu Memristive Devices (Online Talk)	YES	
	Oral 25: ASP 4 Sanjay Kumar, Magnetic Reconnection at Hyperbolic Flux Tube associated with a Confined Flare in NOAA Active Region 12268 (12 Mins + 3 Min)				Oral 27: PTA 4 Dipanjana Hazra Nayak, Numerical Study of Self-focusing and Guiding of q-Gaussian Short (L<p) Laser Pulse in Under Dense Plasma and Effect of Upward Density Ramp (12 Mins + 3 Min)		
	Oral 26: ASP 5 Atherv Saxena, Electron Acoustic Shock And Solitary Waves In Spin Polarized Dense Rotating Quantum Plasma (12 Mins + 3 Min Discussion)				Oral 28: PTA 5 Sankaran K. J, Microplasma Illumination Characteristics of Electrically Conducting Nitrogen-Ion Implanted Boron-doped diamond films (12 Mins + 3 Min)		
4:05 PM to 5:20 PM	Poster Session-III PTA_P_01 to PTA_P_82+ TMS_21 to TMS_P_58 (Total 120) (With a Tea Break in Parallel)					LHC Girst Floor	
5:20 PM to 7:00 PM	Buti Award Presentation Session					LHC-CR-211	
7:00 PM to 9:00 PM	Cultral event followed by Conference Gala dinner					LHC, Second & Groud Floor	
Day-3 (29-12-2025, Monday) at LHC, IIT Tirupati							
Time	Session 7A Astrophysical Plasmas Session Chair: Prof. Ramesh Narayanan, IIT Delhi	Lecture Recording	Room	Session-7	Session 7B Emerging Technologies of plasma Session Chair: Prof. Pramod Gopinath, CUSAT, Cochin	Lecture Recording	Room
9:00 AM to 11:05 AM (30+30+30+15+15+05 (Buffer)) (125 Minutes)	Talk 1: Prof. Ram Prasad Prajapati, JNU, New Delhi (30 Mins) Title: Kinetic and Fluid Descriptions of Jeans Instability In Collisionless Kappa-Distributed Suprathermal Plasmas	YES	LHC-CR-101		Talk 1: Prof. Lakshminarayana Rao, IISc Bengaluru (30 Mins) Title: Plasma activated water and it's application in medicine and agriculture	YES	LHC-CR-211
	Talk 2: Prof. Nishant K. Singh, IUCAA, Pune (30 Mins) Title: Solar magnetic fields and their effects on the surface gravity mode	YES			Talk 2: Dr. Shashi Ranjan, Faraday Earth Pvt. Ltd (30 Mins) Title: RGA Reactor for Nitrogen fixation	YES	
	Talk 3: Dr. Kirit Makwana, IIT Hyderabad (30 Mins) Title: Wave modes and intermittency in plasma turbulence of the solar wind	YES			Talk 3: Dr. Venu Anand, NIT Calicut (30 Mins) Title: Cold Atmospheric Plasma- Generation to Applications	YES	
	Talk 4: Dr. Rajib Kar, BARC, Mumbai (20 Mins) Title: Cold Plasma for Solid Radioactive Waste Treatment	YES			Oral 30: PTA 6 Pushparaj Gandhi Tadavarthi, Sustainable Cold Plasma Systems for Aflatoxin B1 Degradation: Reactor Configurations and Food Quality (12 Mins + 3 Min)		
	Oral 29: ASP 6, Swarniv Chandra, Solar Storms and Their Impact on Space Weather Systems: An Analytical Model Based on High-Volume Space Mission Data (12 Mins + 3 Min)				Oral 31: PTA 7 Ram Prakash Lamba, Design and Simulation of a Cold Cathode Pseudospark Switch for High Repetition Rate Applications (15 Mins + 3 Min)		
11:10 AM to 11:30 AM	Tea- Break						
Time	Session 8A Multi-disciplinary Topics Session Chair: Prof. Abhishek Jha, IIT Tirupati	Lecture Recording	Room	Session-8	Session 8B Cold Plasma Applications Session Chair:	Lecture Recording	Room
11:30 AM to 1:00 PM (30+30+20+10(Buffer)) (105 Minutes)	Talk 1: Dr. Nishant Sirse, IPS Academy, Indore (30 Mins) Title: Tailoring and controlling plasma properties in radio frequency capacitive discharges for semiconductor processing applications	YES	LHC-CR-101		Talk 1: Dr. Trivikram Nallamilli, IIT Tirupati (30 Mins) Title: Role of cold plasma in improvement of techno functional properties of plant proteins	YES	LHC-CR-211
	Talk 2: Dr. Vishant Gahlaut, Banasthali University (30 Mins) Title: Critical Aspects on Thermal Management of TWTs for Space Applications	NO			Talk 2: Prof. Subrahmanyam, IIT Hyderabad (30 Mins) Title: (Plasma chemistry) Plasma Catalysis for CO2 Conversion to Value-Added Products	YES	
	Talk 3: Open Slot				Talk 3: Rajendra Nath Reddy: Renuka Bio Farm, Emerging concern of Agricuture & Food Sector (20 Mins)	YES	
01:00 PM to 2:00 PM	Lunch Break						
2:00 PM to 3:00 PM	Seassion-9 Session Chair: Prof. Arun Sarma, Director-General, Nectar Jaidutt Saraswati Sodha PSSI Plasma Award Talk (30 Min), Parvez Guzdar Award Talk (30 Min)					LHC-CR-211	
3:00 PM to 4:00 PM	Panel Discussion, PSSI GBM, Award distribution and concluding Session						
4:00 PM	Conference Closure						